

ParaEMT EMT-validation campaign (IAS2026 / TX4)

V3 report — instrument qualification

Branch `research/parent-emt-validation` (not pushed)

2026-09-12

Status: V3 STOPPED at gate E0 (instrument qualification).

- **E0, synthetic part** (blind, seed 20260912, 140 cases): **FAIL on one preregistered criterion.**
 - **A5:** 14 of 15 single-channel SG-like ringdowns were resolved, against a requirement of $\geq 95\%$ (15/15).
 - **All other criteria pass:** no crash; **zero wrong-sign verdicts**; 98.3% correct conclusive in S1/S4 and 100% in S2; p95 errors 0.0037 s^{-1} and 0.00086 Hz ; all four DV2-2 regression cases.
- **Not run (stop rule):**
 - the E0 phasor-TDS holdout T1–T9 (still blind);
 - every ParaEMT device gate (B-SG38, B1-GFL35 and B2-GFL37 still blind);
 - EMT04–EMT10.
- No ParaEMT simulation was run in V3.

Contents

1	History (binding)	1
2	V3-0: audit of the frozen v1 estimator (commit 4e35164e)	2
3	The V3 instrument (prereg 6d339006; amendment V3-AM1)	2
4	Gate E0: blind synthetic qualification	2
5	What was not run, and why	3
6	Incidents and disclosure	3
7	Claim ledger	3
8	For the author (decision memo; nothing executed)	4
	What EMT did and did not validate	4

1 History (binding)

campaign	preregistration	outcome (permanent)
V1	c2947bd8 (+ A1 9d06436a)	G1 PASS; G3 FAIL, G4 FAIL (interface / device gates); EMT04–EMT18 BLOCKED
V2	0ab8efd4	SW PASS; G3a FAIL under its preregistered rule (frozen v1 estimator); G3b/G4a/G4b NOT RUN; EMT04–EMT18 BLOCKED
V3	6d339006 (+ AM1 in f8e0d44f)	E0 FAIL (synthetic criterion A5); everything after E0 NOT RUN

V1 and V2 failed their preregistered gates. Subsequent diagnostics identified independent interface and estimator defects. V3 addressed the estimator defect prospectively with a new, independently implemented instrument. On its blind qualification set the instrument made no wrong-sign call and met the accuracy criteria, but it failed one preregistered resolution criterion: single-channel ringdowns, 14/15 resolved against a 15/15 requirement. By rule V3 therefore stopped before any ParaEMT device or portfolio run.

2 V3-0: audit of the frozen v1 estimator (commit 4e35164e)

emt_estimator.py was not modified.

- **Unresolved everywhere.** resolved=False in all four V2 G3a evaluations, with a residual of about 0.999. The G3a rule compared α and f unconditionally.
- **B-SG36.** The EMT run and its reference share the same spurious $+6.3\text{ s}^{-1}$ mode.
- **DV2-2.** Three LinAlgError crashes and one unresolved case.
- **Mechanism:**
 - spurious $|z|$ up to 18.9;
 - the full-record Vandermonde z^k spans 10^{26} – 10^{95} , or overflows (10^{355});
 - lstsq keeps rank 1–3 of 10–16;
 - the $\exp(2\alpha T)$ energy weight (10^{43}) selects a spurious mode with a residue of 10^{-22} .

3 The V3 instrument (prereg 6d339006; amendment V3-AM1)

Estimator. A new multi-channel bounded variable-projection damped-mode estimator.

- **Model.** For each channel, one damped mode plus an affine term, with time centred on the window.
- **Data.** 0.1-s block means, which leave every exponential's s unchanged.
- **Fit.** One Householder QR of a common 4-column design matrix. Only (α, f) are optimized, within bounds.
- **Uncertainty and decision.**
 - channel and time-block jackknife CI;
 - resolved flag R1–R4;
 - verdict from the CI with $\text{EPS} = 0.002$;
 - targeted tracker (committed $f_{\text{pred}} \pm 0.15\text{ Hz}$; α_{pred} never used);
 - broad-band sentinel;
 - adaptive run length $30 \rightarrow 60 \rightarrow 90\text{ s}$.
- **Removed.** There is no full-record Vandermonde and no exponential energy weighting.

V3-AM1, committed *before* any blind data existed.

- **Problem.** On the development data (seed 1111, DV2-2, the non-holdout TDS P4 BASE), the unweighted fit carried a nuisance-mode bias of 0.0055 on DV2-2. It also left one strongly damped S6 development case unresolved.
- **Change.** An envelope-equalizing weight $\exp(-\alpha_w \tau)$, with $|\alpha_w| \leq 3/L$, taken from a first unweighted pass and then fixed. The optimization is still over (α, f) only, and no criterion was changed.
- **Development result:**
 - DV2-2 bias 0.00036;
 - S6 5/5 on the development set;
 - development TDS $\Delta\alpha = 5.6 \times 10^{-4}$ against the linear eigenvalue.

4 Gate E0: blind synthetic qualification

stratum	n	resolved	correct conclusive	wrong	p95 $ \Delta\alpha $	p95 $ \Delta f $
S1 single mode	40	40/40	39/40 (miss: $\alpha = +0.0029$, UNRESOLVED; not in A3)	0	9.5×10^{-7}	1.7×10^{-7}
S2 target + nuisance	25	25/25	25/25	0	3.4×10^{-3}	9.5×10^{-4}
S3 near boundary	20	20/20	12/20 (rest UNRESOLVED, allowed)	0	3.6×10^{-8}	7.7×10^{-9}
S4 heterogeneous	20	19/20	19/20	0	2.2×10^{-3}	4.7×10^{-4}
S5 two close modes	20	17/20	17/20	0	4.9×10^{-3}	1.3×10^{-3}
S6 single-channel ringdown	15	14/15	14/15	0	5.8×10^{-3}	8.9×10^{-4}

criterion	rule	result	verdict
A1	no crash	none	pass
A2	0 wrong-sign verdicts, $ \alpha \geq 0.005$	0	pass
A3	S1+S4, $ \alpha \geq 0.01$: $\geq 95\%$ correct conclusive	57/58 = 98.3 %	pass
A3b	S2, $ \alpha \geq 0.01$: $\geq 80\%$	100 %	pass
A4	p95 $ \Delta\alpha \leq 0.005$, p95 $ \Delta f \leq 0.005$	0.0037 s ⁻¹ ; 0.00086 Hz	pass
A5	S6 $\geq 95\%$ resolved	14/15 = 93.3 %	FAIL
A6	DV2-2 regression	4/4	pass

Reported: the CI coverage of α_{true} is 99.3%; S3 and S5 resolution is 100% and 85%; the sentinel flagged nothing.

The failing case. S6 #137 has $\alpha_{\text{true}} = -0.473 \text{ s}^{-1}$ and $f = 1.067 \text{ Hz}$, a single channel with a slow real mode superposed, in the window [1.5, 10] s.

- The estimate was $\hat{\alpha} = -0.455$.
- The time-block jackknife CI half-width was 0.027 s^{-1} , above the 0.02 limit (R4), with a median residual of 0.26.
- The instrument reported **UNRESOLVED instead of a biased value**.
- The criterion nevertheless requires all 15 cases to resolve.

The unresolved S4 and S5 cases also had biased point estimates (e.g. -0.419 against -0.262), and each was withheld as UNRESOLVED, never reported as a value.

Gate E0: FAIL. V3 stops (prereg V3 §1.3). No other estimator is tried in this campaign.

5 What was not run, and why

item	status	consequence
E0 phasor-TDS holdout T1–T9	NOT RUN (E0 failed on the synthetic part)	the traces were never generated, so the set stays blind
G3a, G3b (R-SG30, R-SG36, B-SG38)	NOT RUN	B-SG38 has never been simulated
G4a, G4b (R-GFL30, B1-GFL35, B2-GFL37)	NOT RUN	the GFL holdouts are still blind; the V2 voltage interface has never run in a network
EMT04, EMT05, EMT06, EMT07/08, EMT10	NOT RUN	no IEEE-39 EMT portfolio result exists

Prepared but never executed (committed separately and labelled): `EMTV3_refs.py`, `EMTV3_gates.py`, `v3_case.py`, `EMTV3_science.py`, `overlay/tx4_empt_v3.py`. The gate and science scripts assert that the preceding gates passed, so they cannot run under the V3 record.

6 Incidents and disclosure

1. **The first blind E0 execution was killed** by the system for low memory, mid-way through S5. The cause was system-wide memory pressure from unrelated applications. No other process was touched.
 - Its progress log showed *verdict labels* (not true α values) for cases up to S5 #6.
 - The code was not changed. E0 was re-executed with the identical code and seed.
2. **Determinism.** The E0 rerun required by prereg §8 is recorded in `results/EMTV3/E0/E0_rerun_identity.json`.
3. **Commit order.** The implementation commit `f8e0d44f` preceded all blind data. The style-only commit `70d5be8e` carries a byte-order-mark character in its subject line (cosmetic).

7 Claim ledger

No change. No claim receives EMT support, and none is EMT-refuted.

- V19 remains the only EMT QUANTITATIVELY REPRODUCED claim.
- V20 and V21 keep their final-ledger status, which rests on the independent ANDES reproductions.

8 For the author (decision memo; nothing executed)

What V3 established about the instrument. The new VP estimator does not crash, gives no wrong-sign verdict, and is accurate to $p95\ 0.004s^{-1}$. Where it cannot separate modes, it withholds a value rather than report a biased one. This is instrument evidence only.

Why V3 stopped. The preregistered criterion demanded that all 15 single-channel ringdowns resolve, and one heavily damped case did not.

Options.

- **(a) Close the EMT line with its present record.** The network and operating point are EMT-reproduced (V19). Every other claim keeps its phasor and ANDES status. This is consistent with “no V4 merely to obtain agreement”.
- **(b) A new, separately justified campaign,** with its own preregistration. Its blind elements would be the unused TDS holdout T1–T9, B-SG38, B1-GFL35, B2-GFL37 and a fresh synthetic seed.

What EMT did and did not validate (after V1, V2 and V3)

The table separates **theorem validity** from **model-fidelity corroboration**.

Theory / claim	Mathematical status	Phasor evidence	Independent phasor evidence	EMT evidence (V1+V2+V3)	Final status
V01, V02, V08, V22, V26, T04, T06, T08 (theorems)	proved	yes	where applicable	not applicable	PROVED
V03 P4 trap, $\kappa = 4$ [B02]	—	PCV02; G2 TDS	ANDES Phase 8, 32/32	BLOCKED / NOT RUN	VALIDATED (model-specific witness)
V06, V07 policy dependence [B01]	structure proved	clean g -only counterfactual	ANDES Phase 8	BLOCKED / NOT RUN	VALIDATED
V10, V11 port derivative, line ranking	proved (V10)	PCV04	ANDES R3 11/12	BLOCKED / NOT RUN	PROVED + VALIDATED; VALIDATED (local)
V13, V14 remediation, governors	—	PCV04; FC03; G2	partial ANDES	BLOCKED / NOT RUN	VALIDATED (model-specific); BENCHMARK-SPECIFIC
V15–V18; V30, B07, B10	— / normal form	PCV05; G2 TDS	—	BLOCKED / NOT RUN	as in the ledger
V19 network and operating point [I01]	—	reference	ANDES; pandapower after the tap fix	EMT QUANTITATIVELY REPRODUCED (V1 G1)	CONDITIONAL (pandapower translation)
V20 SG dynamics [I02, I03]	—	internal	ANDES R2/R3, E1	V1 G3 FAIL; V2 G3a FAIL; V3 NOT RUN → EMT UNRESOLVED	VALIDATED (by ANDES)
V21 custom GFL [I04]	—	project DAE	ANDES Phase 8, GATE 5 PASS	V1 G4 FAIL; V2/V3 NOT RUN → EMT UNRESOLVED	VALIDATED as a reproduction of computation; EMT reproduction not achieved
cumulants, closure, V05, V12, B04, B06, negatives	as in the ledger	as in the ledger	—	not applicable / BLOCKED	unchanged